

Examining the Effect of Salt Hydrates, Supporting Materials, and Salt Loadings on Thermochemical Reaction Efficiency for Renewable Energy Storage

Introduction

In many industrial applications, such as data centers, renewable energy can't be used due to its **intermittent nature**. However, thermochemical energy storage systems present a potential solution.

Thermochemical Energy Storage uses salt hydrates to store energy through reversible chemical reactions over long periods of time. **During dehydration**, dry air is introduced, triggering an endothermic reaction that removes water molecules from salt hydrates. This reaction converts thermal energy into stored chemical potential within the anhydrous salt. **During hydration**, wet air is introduced, triggering an exothermic reaction as the salts absorb water and rehydrate, releasing the stored energy.

Existing Work

- 1) Clark et al., 2021
 - Identified promising salt hydrates
 - Solely evaluated hydration kinetics
 - Performed a salt kinetic pure analysis
- 2) Padamurthy et al., 2022
 - Tested four salts sequentially.
 - Supplied Nitrogen gas at a high flow rate
 - Measured percentage of mass lost/gained.
- 3) Ji et al., 2022
 - Investigated 2 salt hydrate combinations
 - Evaluated zeolite and silica gel supports
 - Packed the particles using metal mesh.

Purpose

To study which combinations of salt hydrates, supporting materials, and salt loading percentages optimize long-term thermochemical energy storage.

Hypothesis

The combination of MgSO₄, silica gel, and 60% salt loading will have the highest efficiency.

Independent Variables

Salt Hydrates: CaCl₂, MgSO₄, SrCl₂
Supporting Materials: Silica Gel, Zeolite, Alumina
Salt Loadings: 45%, 60%

Dependent Variable

Thermal Efficiency (energy release over energy input)

Control: 3 Supporting Material Blanks

Methodology: Construction

Construct the Reactor

Create a water bubbler system

Integrate the thermocouples

Connect with silica tubing



Figure 1: Labeled Image of the Reactor

Methodology: Salt Composites

Use **incident wet impregnation** to integrate the salt hydrates into the supporting materials (Galazutdinova et al., 2022).

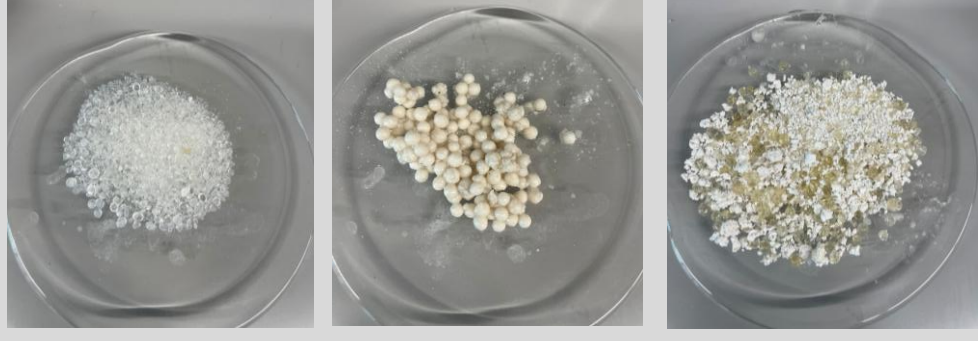


Figure 2: Salt Composites

Methodology: Cycling

Perform **dehydration** cycles using the oven and vacuum chamber.

Perform **hydration** using the water bubbler, reactor, and air tank.

Repeat each cycle 5 times to measure degradation. Replace the composite material. (Miao et al., 2021)

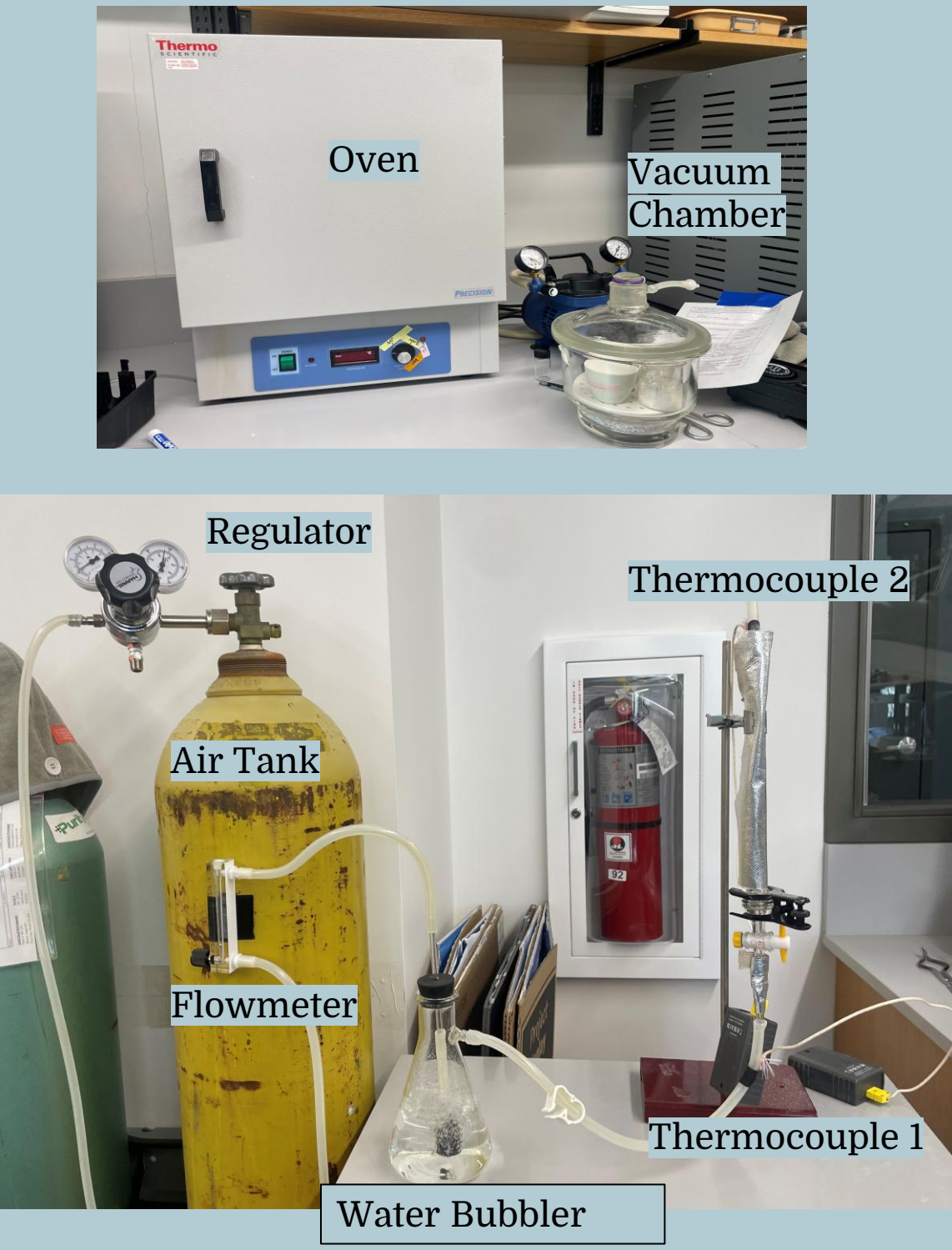


Figure 3: Labeled Image of the Hydration System's Parts

Results & Analysis

Dehydration Data Manipulation:

Chemical Energy (J) = 1000 * nH₂O * ΔH_{dehydration}

Table 1: Processed Dehydration Kinetics for MgSO ₄ Composite						
Trial & Cycle	Initial Mass (g)	Final Mass (g)	Mass Loss (g)	Support Mass Loss (g)	Isolated Water Loss (mol)	Chemical Energy (J)
Trial 1, Cycle 1	9.7000	8.0000	1.7000	0.3220	0.0765	3870.3734
Trial 2, Cycle 1	9.0400	7.6700	1.3700	0.3001	0.0594	3005.0249
Trial 1, Cycle 2	8.3500	7.5400	0.8100	0.2772	0.0296	1496.4567
Trial 2, Cycle 2	9.4000	7.7000	1.7000	0.3187	0.0767	3879.6985
Trial 1, Cycle 3	8.0000	6.9600	1.0400	0.2656	0.0430	2175.1119
Trial 2, Cycle 3	8.1100	6.8800	1.2300	0.2693	0.0533	2698.5206
Trial 1, Cycle 4	7.4000	6.4000	1.0000	0.2457	0.0419	2118.7117
Trial 2, Cycle 4	7.3500	6.6200	0.7300	0.2440	0.0270	1365.0063
Trial 1, Cycle 5	6.8400	6.2800	0.5600	0.2271	0.0185	935.0734
Trial 2, Cycle 5	6.6400	6.2800	0.3600	0.2204	0.0077	391.9695

Hydration Data Manipulation:

A temperature-time graph was generated by Vernier Graphical Analysis for each composite.

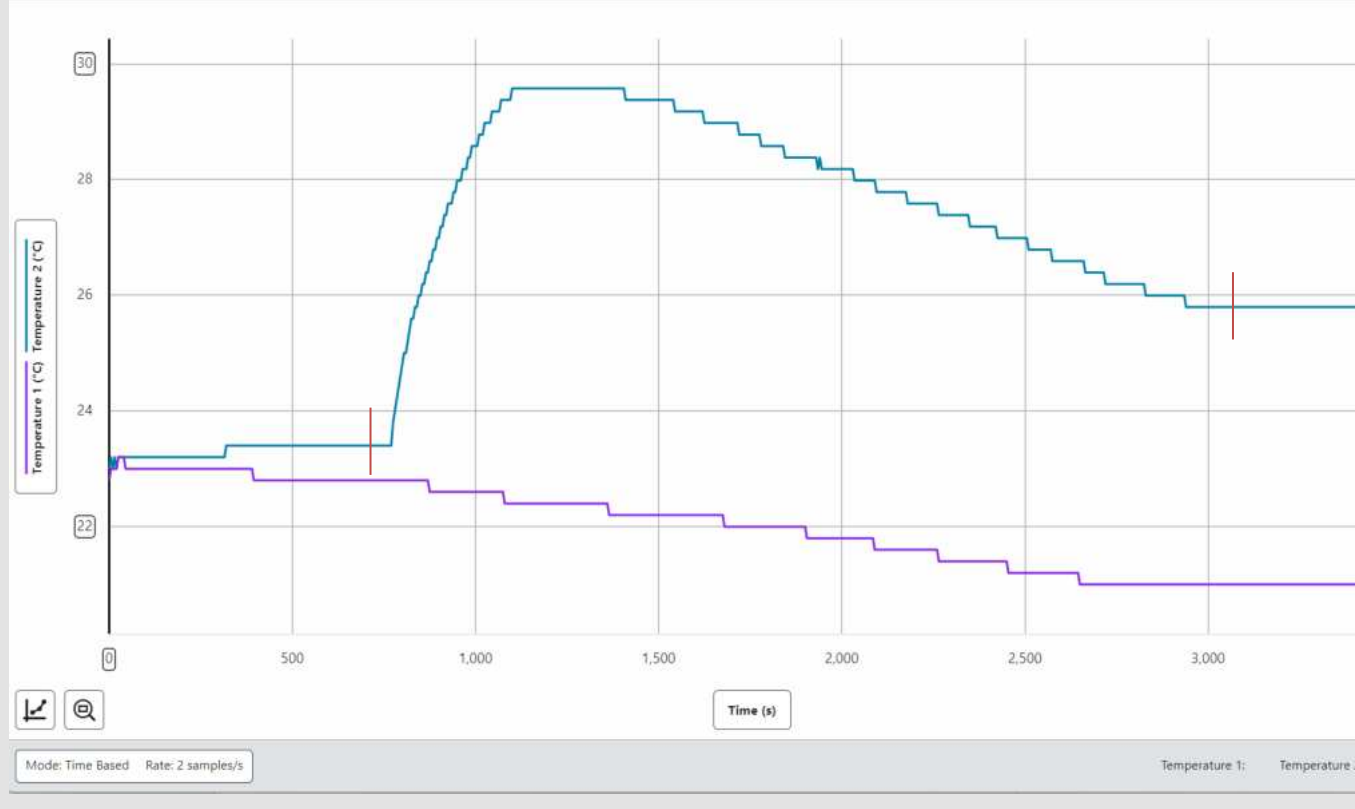


Figure 4: Screenshot of Vernier Generated Temperature-Time Graph

The total area under the curve was calculated using formulas in Excel.

Figure 5: Excel Spreadsheet for One Cycle									
Time (s)	Temperature (°C)	Temperature (°C)	Temperature (°C)	Temperature (°C)	Temperature (°C)	Temperature (°C)	Temperature (°C)	Temperature (°C)	Temperature (°C)
0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
100	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
200	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
300	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
400	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
500	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
600	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
700	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
800	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
900	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1000	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1100	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1200	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1300	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1400	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1500	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1600	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1700	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1800	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
1900	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0
2000	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0	20.0

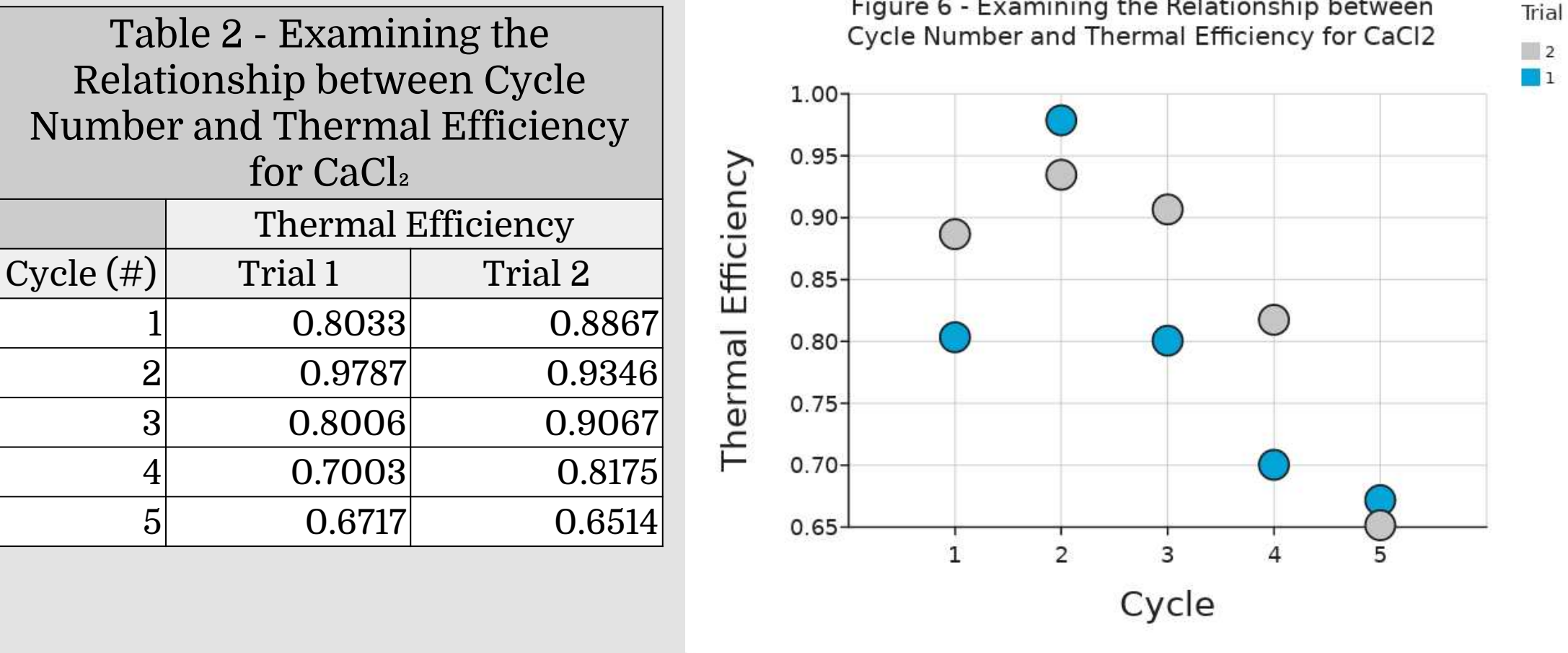
Figure 5: Excel Spreadsheet for One Cycle

Results & Analysis

Thermal Efficiency was calculated by dividing energy released by energy inputted.

$$\eta_{RT} = \frac{Q_{out}}{E_{in}}$$

CaCl₂ x Activated Alumina x 60% Salt Loading Composite



SrCl₂ x Zeolite x 45% Salt Loading Composite

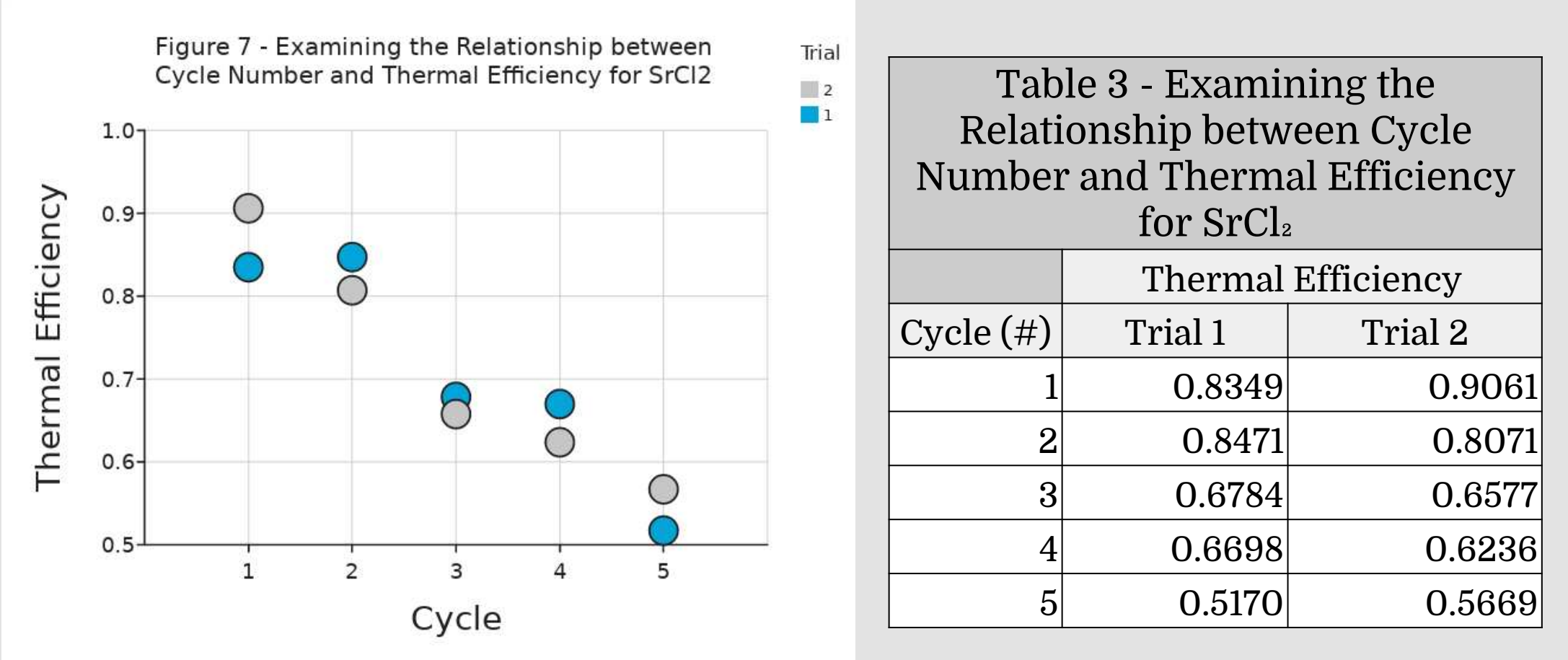
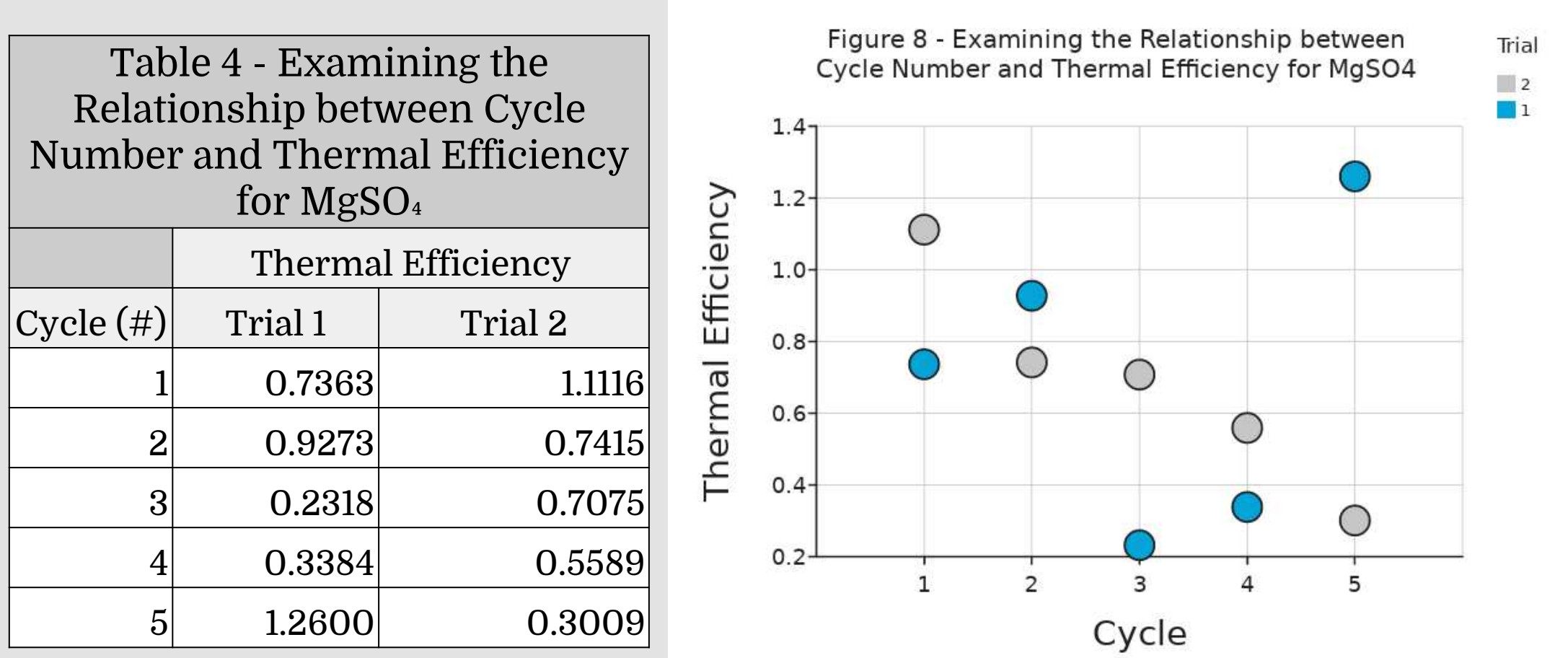
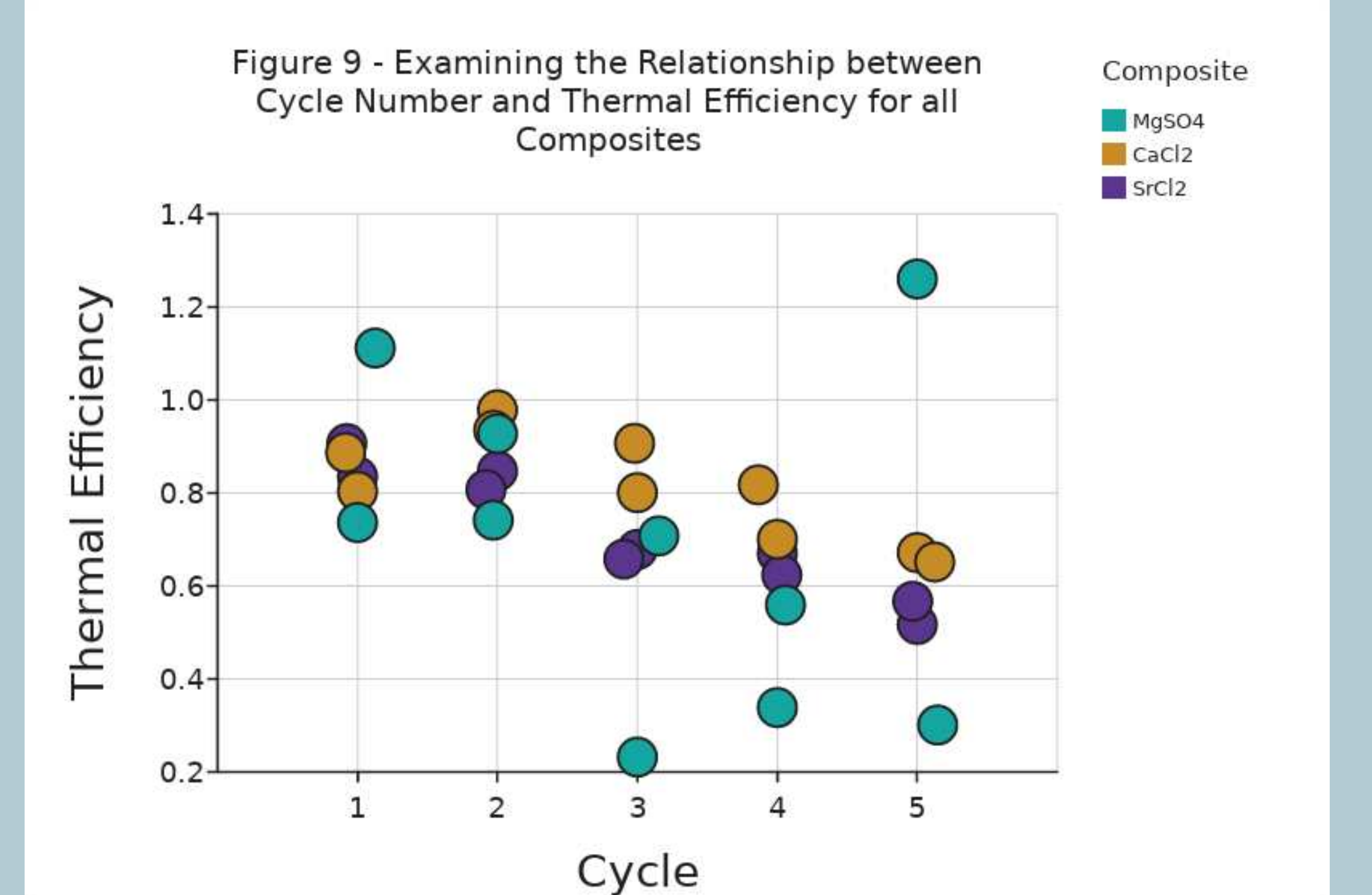


Table 3 - Examining the Relationship between Cycle Number and Thermal Efficiency for SrCl ₂		
Cycle (#)	Thermal Efficiency	
	Trial 1	Trial 2
1	0.8349	0.9061
2	0.8471	0.8071
3	0.6784	0.6577
4	0.6698	0.6236
5	0.5170	0.5669

MgSO₄ x Silica Gel x 60% Salt Loading Composite



The thermal efficiencies were then compiled and placed on the same graph. The thermal efficiency scale starts at 0.2 to show additional detail.



**All graphs, images, and diagrams were created by the presenter.

Results & Analysis

A Kruskal-Wallis test was used to determine if there were statistically significant differences between the thermal efficiencies for the three composites.

Table 5 – Statistical Results of Kruskal-Wallis Test on Thermal Efficiency Data			
Degrees of Freedom (df)	H-Statistic	P-Value	Interpretation of P-Value
8	12	0.30	A P-value of 0.30 means little to no indication that the groups might be different

The p-value was found to be 0.30, which is greater than the statistically significant value of 0.05. Thus, we fail to reject the null hypothesis. Based on the results, a post-hoc test was completed.

Results of post-hoc Dunn's Test (with Bonferroni correction):		
Pair	P-value	Interpretation of P
SrCl2 - CaCl2	0.46	Means: little to no indication that the groups might be different
SrCl2 - MgSO4	1.00	Means: that the groups are NOT different
CaCl2 - MgSO4	0.64	Means: no evidence that the groups might be different

Conclusion

- The objective of this project was reached as the thermal performance of three composites was compared across two trials and five dehydration-hydration cycles.
- The original hypothesis was not supported. The CaCl₂ activated alumina performed the best overall, followed by the SrCl₂ zeolite and then the MgSO₄ silica gel. CaCl₂ activated alumina had the most consistent thermal efficiency values from 0.65 to 0.98.
- The Kruskal-Wallis test found that the differences between composites were not statistically significant, with a p-value of 0.30. The lack of statistical significance was likely due to the small sample size.
- One source of error was airflow rate, since Q_{out} calculations depended on the 10 L/min flow rate.
- Another limitation was that dehydrated salts were sometimes stored overnight in a vacuum chamber.
- Future research should examine more cycles, trials, and combinations of salts, supports, and loadings.
- This project suggests that **CaCl₂ activated alumina at 60% loading** may be a promising composite for thermochemical energy storage, while also showing that experimental performance depends on more than theoretical energy density alone.

Sources

1. Blijlevens, M., Mazur, N., Kooijman, W., Fischer, H., & Huinink, H. (2022). A study of the hydration and dehydration transitions of SrCl₂ hydrates for use in heat storage. *Solar Energy Materials and Solar Cells*, 242. <https://doi.org/10.1016/j.solmat.2022.111770>
2. Chen, X., Zhang, Z., Qi, C., Ling, X., & Peng, H. (2018). State of the art on the high-temperature thermochemical energy storage systems. *Energy Conversion and Management*, 177, 792-815. <https://doi.org/10.1016/j.enconman.2018.10.011>
3. Clark, R.-J., & Farid, M. (2021, October). Hydration reaction kinetics of SrCl₂ and SrCl₂-cement composite material for thermochemical energy storage. *Solar Energy Materials and Solar Cells*, 231. <https://doi.org/10.1016/j.solmat.2021.111311>
4. Donkers, P., Sogutoglu, L., Huinink, H., Fischer, H., & Adan, O. (2017). A review of salt hydrates for seasonal heat storage in domestic applications. *Applied Energy*, 199. <https://doi.org/10.1016/j.apenergy.2017.04.080>
5. Galazutdinova, Y., Clark, R.-J., Al-Hallaj, S., Kaur, S., & Farid, M. (2024). New thermochemical salt hydrate system for energy storage in buildings. *Energies*, 17(20). <https://doi.org/10.3390/en17205228>
6. Humbert, G., & Sciacovelli, A. (2023). Topology optimization for mass transfer enhancement in open thermochemical energy storage reactors. *Journal of Energy Storage*, 64. <https://doi.org/10.1016/j.est.2023.107132>
7. Ji, W., Zhang, H., Liu, S., Wang, Z., & Deng, S. (2022, July). An experimental study on the binary hydrated salt composite zeolite for improving thermochemical energy storage performance. *Renewable Energy*, 194, 1163-1173. <https://doi.org/10.1016/j.renene.2022.06.024>
8. Miao, Q., Zhang, Y., Jia, X., Li, Z., Tan, L., & Ding, Y. (2021). MgSO₄-expanded graphite composites for mass and heat transfer enhancement of thermochemical energy storage. *Solar Energy*, 220. <https://doi.org/10.1016/j.solener.2021.03.008>
9. Padamurthy, A., Nandanavanam, J., & Rajagopalan, P. (2022). Assessment of selected salt hydrates for thermochemical energy storage applications. *Materials Today Proceedings*, 64 (4), 1609-1615. <https://doi.org/10.1016/j.matpr.2022.04.048>
10. Wang, H., Yuan, X., Zhang, K., Lang, X., Chen, H., & Yu, H. (2024). Performance evaluation and optimization of data center servers using single-phase immersion cooling. *International Journal of Heat and Mass Transfer*, 221. <https://doi.org/10.1016/j.ijheatmasstransfer.2023.125057>